

MedZen Speech Platform

Project Handoff Report — Claude (sole implementation engineer + reviewer)

Date: 2026-08-20 **Repo:** ~/Documents/medzen-platform (github.com/fotso94/medzen-platform, private) **HEAD:** 6d7d68c

AWS account: 558069890522 **Region:** eu-central-1 (all resources)

Purpose: allow another AI assistant or senior engineer to independently understand, audit and continue this project without prior conversations. Written for a follow-up independent audit by ChatGPT.

Method: every load-bearing claim below was re-verified live on 2026-08-20 against git, the full test suite, S3, ECR, EC2, SageMaker, IAM, SSM, Secrets Manager, EKS and Cost Explorer — not reproduced from memory. Claims that could not be re-verified are explicitly marked NOT VALIDATED. This verification pass itself found and fixed one latent deploy-day bug (§11, item 8).

Read this first: §19 HANDOFF — START HERE (last page) is the entry point for the next engineer. §3 is the one-page status dashboard.

1. Project objective and original architecture

MedZen is building a **medical-assistant speech platform for African languages**: a patient speaks in their language; the platform transcribes (ASR), retrieves grounded medical guidance (RAG), generates a cited answer (LLM), and speaks it back (TTS). Target: ~20 in-scope languages for generation, 40+ languages with frozen evaluation sets.

The plan is phased: **B1** contracts & schemas → **B2** data ingest (A3 manifest standard) → **B3** per-language baselines → **B4** frozen evaluation suite → **B5** training (fine-tuned ASR per language) → **B6** serving (5-service platform on EKS) → **B7** CI/CD activation. Base ASR model: Meta omniASR (CTC 1B family). LLM: AWS Bedrock. TTS: Fish Audio (bought voices) until in-house TTS data exists.

Current architecture (all five service images built and pushed; not yet deployed): client → **speech-orchestrator** (API-key auth via Secrets Manager, SSM registry routing) → **asr-runtime** (omniASR + approved fine-tuned weights from S3) → **rag-index** → **llm-gateway** (Bedrock Claude Haiku 4.5, citation-bound JSON contract) → **speech-tts-gateway** (Fish Audio, per-language voice registry in SSM). Every gateway defaults to an offline fake mode; real modes are explicit env opt-ins; unknown modes refuse (fail-closed).

2. Governance: who built what, and when Claude took control

Until 2026-08-14 the project was implemented by **ChatGPT/Codex** with **Claude as independent reviewer** (per-service exit reviews, PASS/CONCERNS/HOLD verdicts). On **2026-08-15** the owner transferred the implementation role: Claude became **sole engineer + self-reviewer**.

Evidence: ~/Documents/medzen-shared/codex_status.txt line 1: "Report ID: CLAUDE-ENGINEER-STATUS-2026-08-15-001 ... Engineer: Claude (sole engineer + self-reviewer per owner role transition 2026-08-15)". Git: 756 commits dated ≤2026-08-14, 320 commits since 2026-08-15 (total 1,074+ since 2026-07-28).

Standing owner constraints that bind any successor:

- Signed data agreements (LIC-2026-002 ShareAlike, LIC-2026-003 africanvoices, LIC-2026-004 NaijaVoices) are **PRIVATE with the owner** — repo carries attestation records only; never commit or upload the papers.
- The live ECS service **medzen-tts-dev** (cluster medzen-tts-dev, service medzen-tts-gateway) is the working reference/fallback — **never modify it**; review only.
- The engineering agent never handles credentials (HF login and API keys are loaded by the owner directly).
- Cost reporting: on-demand pricing as basis; add \$100 to every reported cost maximum.
- Training spend requires an owner-approved packet: bindings JSON → render → validate → written authorization phrase in the shared file → launch. Job names are single-use.

3. Current phase and overall status (2026-08-20, ~03:45 UTC)

Area	Status	Evidence (verified 2026-08-20)
Test suite	2,202 passed, 1 skipped, 7 deselected (92 s)	pytest run 2026-08-20 on HEAD
Shipped model	kinyarwanda ASR v1 APPROVED: 0.372 WER / 0.089 CER	s3://medzen-speech/eval/approveshinyarwanda-asr-v1/model-pr (E55 Gp)
Serving images	All 5 platform images in ECR, provenance-stamped, contained ECN ingest to B5SDP NOT-RepN/A0KXSlender; eksctl 0/4, ek	ECN ingest to B5SDP NOT-RepN/A0KXSlender; eksctl 0/4, ek
In flight 1	kinyarwanda 2,000 h re-pull: decode DONE, uploading 378,700 x 1,024, 62860p (33%)	../research/b5-training/rw-ingest/host.l
In flight 2	Pidgin (africanvoices): 7/42 archives in S3 (owner Mac uploading 13-d 744B; left) a self-sequencing log if S3CibC Banding 1"	ding 13-d 744B; left) a self-sequencing log if S3CibC Banding 1"
In flight 3	NaijaVoices: hausa COMPLETE (113,439 clips); igbo uploading 25,400,920B3/nyorbaasqueuS8; Mac ingest log	ing 25,400,920B3/nyorbaasqueuS8; Mac ingest log
In flight 4	SageMaker quota 1→2 concurrent g6.xlarge training jobs: CAS643PE60B64WSD6mantiwiev66AE9D73	CAS643PE60B64WSD6mantiwiev66AE9D73
Next gate	kinyarwanda v2 FULL fine-tune on ~2,000 h once re-pull lands skip code 2026-08-16 (over again in kinyarwanda-v1.yaml)	skip code 2026-08-16 (over again in kinyarwanda-v1.yaml)

Phase: B5 (training) mid-campaign + B6 (serving) built-and-proven but dark. Nothing about the serving stack is live; the only running cost sources are the two ingest boxes and the EKS control plane (\$15).

4. What existed before Claude took control (verified, not assumed)

Built in the Codex era (2026-07-28 → 2026-08-14, 756 commits), verified present and working today:

- **A3 manifest standard + ingest pipeline core** (pipeline/ingest.py, pipeline/adapters/base.py): checksum-verified rows, licence tiers, speaker/text-disjoint splits, byte-dup refusal, near-dup validator.
- **Adapters:** soreva (frozen eval sets), waxalnlp, aaf, lrsc, yemba_egra, meta_omnilingual, kallaama, common_voice (base).
- **Frozen evaluation pools** under s3://medzen-speech/eval/ for 40+ languages (SOREVA et al.) — these are the benchmark; training data must never leak into them.
- **Baseline machinery** (scripts/run_baseline.py, paired decode comparison) and language registry (registry/languages/*.yaml, 20 generation languages).
- **B6a runtime images:** medzen-asr-runtime:b6a-003c-89f94d3, medzen-model-loader:b6a-003b-69bdce1 (ECR, 2026-08-04).
- **asr-eval-runtime image** + K8s Job eval harness on EKS medzen-speech — eval attempts 1–25 (attempt 25 failed; root cause was found by Claude, §11).
- **Contracts:** platform/contracts/speech-v1.yaml (adopted), llm-v1, tts-v1 (local-mock era), services.yaml, fixtures + JSON schemas.
- Original trainer image medzen-trainer (2026-08-01, superseded).

Evidence: git log --until=2026-08-14; ls pipeline/adapters; aws ecr describe-images (digests §13); aws s3 ls s3://medzen-speech/eval/ (40+ language prefixes).

5. What Claude implemented after taking control (320 commits, chronological)

When	Arc	What was delivered
08-15	Eval unblocked	Attempt-25 root cause (asset-card absolute /models paths not mounted) + fix + render-time mount validator; attempts 26
08-16→17	Full evaluation suite	10 frozen shards, attempts 29–44, one attempt per authorization revision; complete zero-shot baseline pool for the 40+ e
08-17	Training stack	pipeline/omniasr_train.py (LoRA, CTC scope, env-dict config, refusal on unknown modes) + scripts/b5_sagemaker_job.p
08-17→18	Wave-1 + T6 + forensics	gb3 multilingual shared-adapter training (r2) → T6 gate on frozen pools: 0/9 pass, 7/9 REGRESSED → full forensics: lr 1
08-18→19	Kinyarwanda arc	Per-language probes (clean U-curve 0.375@1800 steps) → decisive 300 h rank-32 run (dev plateau 0.355) → T6-style fr
08-18→19	Data campaign	Tier A ingest (10 corpora: CV hausa/igbo/yoruba/english-capped, FLEURS train, WaxalnLP malagasy, NCHLT sepedi +
08-19	Full fine-tune prep	MEDZEN_TRAIN_MODE=full in trainer (full-checkpoint save/load, all-params grad); ft image medzen-trainer-omniasr:ft-2
08-19→20	B6 services real	llm-gateway: BedrockProvider (Converse API, eu.anthropic.claude-haiku-4-5 inference profile, JSON contract, cited-ids g
08-20	Pidgin prep + this handoff	africanvoices adapter (XLSX quirk, fail-closed label checks) + self-sequencing ingest box; SageMaker quota request 1→

Evidence: git log --since=2026-08-15 (320 commits); decision/evidence records in platform/decisions (270 files) and platform/evidence (379 files); codex_status.txt entries 2026-08-15 → 2026-08-20.

6. What Claude modified or corrected from previous work

Previous state (Codex era)	Correction + why
Eval image asset cards resolved checkpoints at absolute /models paths, all of models mounted + appended time at 25 failed in use 2 any workload whose card	explicit max_new_tokens; image rebuilt + republished
Whisper decode capped by default max_new_tokens → truncated	fixed root cause, not skipped
5 pre-existing red tests (A3 drift, registry, F placeholder, time-bomb	chunked readback + consumer audit
SSM aggregate readback exceeded parameter size on big shards	abandoned on T6 evidence (0/9, interference is structural); replaced by per-language str
Shared multilingual adapter as the B5 training shape	per-media type noisedives from the provider (own deep-review catch before any real call v
tts-gateway validated audio against a synthetic constant media type	role seg in medzen/registry/tts/voices (inside existing grant)
Voice registry SSM path /medzen/tts/voices sat OUTSIDE the pod	inflight by his report's verification in default 20e medzen/speech/fish-api-key, runbook corre
Fish key secret name medzen/fish-api-key (does not exist) baked	

7. What has been tested and independently verified

Claim	Verification evidence
Repo is healthy at HEAD 6d7d68c	full suite 2,202 passed / 1 skipped / 7 deselected, 92 s, 2026-08-20
kinyarwanda v1 beats baseline on the FROZEN poggate run on ephemeral g6.xlarge eval box, pinned image medzen-asr-eval-runtime@sha256:dafc4b14...	approved/kinyarwanda/asr/v1/model.pt (1,951,673,613 bytes, 2026-08-19) = merged rank-32 step-6000
v1 artifact is the gated artifact	ECR digests listed §13, SOURCE_COMMIT build-args enforced by Dockerfiles; verified via describe-ima
All 5 serving images exist, provenance-stamped	b6_live_contract_probe.py --full inside the pushed image: PASS (file + streaming), fixtures/sibling packa
Orchestrator container honours the live contract	unit contract tests (stub client) + live Converse smoke during build (eu inference profile requirement disc
Bedrock provider maps the contract	TIER-A-REVIEW-2026-001: checksums re-verified from S3, licence labels vs sources, zero byte-leaks v
Tier A corpora are sound	ingest EXIT 0 2026-08-19T23:50:46Z, 113,439 clips ≈100 h, manifest in curated/hausa/asr/nv_hausa/v1
NaijaVoices hausa corpus complete	unit tests (2) + dry run against real Batch_1 rows on the Mac: 5 records pass all A3 checks
africanvoices adapter reads the real archives	

8. What exists but has NOT been fully validated

- **Serving deploy**: no image has ever run on the EKS cluster as a service (node groups at 0). Container proof ≠ cluster proof: IRSA/roles, ALB, registry namespace, secrets wiring are untested end-to-end.
- **Fish real synthesis via the platform gateway**: the real provider code is reviewed + unit-tested; no end-to-end synthesis has run because the platform Fish secret is EMPTY (owner action, §16).
- **Full fine-tune mode on real GPU at scale**: unit-tested and image-built (ft-2026-08-19a); never yet run as a SageMaker job. The v2 job will be its first live use — watch checkpoint sizes and save cadence.
- **orchestrator deployed_http_ssm mode** against real cluster services (only fixture mode has run end-to-end).
- **In-flight corpora** (rw v2 2,000 h, NV igbo/yoruba, Pidgin): not yet curated/reviewed — do not train on them until the Tier-B review and gb5 version binding exist.
- **LLM answer quality per language**: gateway contract is enforced, but the ~\$5 multilingual calibration run (task #36) hasn't happened.
- **akan/serer baselines** are known-wrong (label-mapping mismatch, task #35) — do not gate anything on them.

9. Remaining tasks in recommended execution order

#	Task	Trigger / dependency
1	rw 2,000 h upload completes → curate v2 (drop_both stats, byte-leak strip vs cv17-testsuite on box) → Ogb7 decision on bidway	after checkpoint v2 (2000) h relogian n 98%
2	v2 FULL fine-tune packet (ft image, MEDZEN_TRAIN_MODE=full, lr ~1e-5, ~40k steps)	check 00s SAGEVA pidgin 2 days
3	Pidgin ingest completes → Tier-B deep review (NV + AV: checksums, licences, leak boxes)	inter (34w/24h) app 51% AWS spend
4	Next-wave decision table to owner: pidgin/hausa/igbo/yoruba proposed bars vs baseline	inshape verdict
5	akan/serer eval-baseline label fix (#35)	independent, small
6	LLM multilingual calibration ~\$5 (#36)	independent, small
7	Serving deploy per B6-DEPLOY-RUNBOOK (#37): owner operating-shape decision (always decision 200 required)	max incl. +\$100 rule / business-h
8	Wave-2 re-scope under per-language strategy (13 approved languages superseded	

10. Training / evaluation experiments — results and conclusions

Experiment	Result	Conclusion
T5 calibration (yemba, CTC head)	PASS, <\$10	trainer + packet flow proven
gb3 wave-1 shared adapter (9 langs, r2)	T6: 0/9 pass, 7/9 regressed vs zero-shot	NO_PROMOTION (B5-T6-GATE-2026-001)
Weight-delta forensics (807 tensors)	non-target drift 0.17%; q/v median 21.6%, max 121%	merge machinery correct; lr 1e-3 runaway is the cause
Checkpoint sweep (r2)	damage present from step 400	no salvageable checkpoint (B5-SWEEP-2026-001)
Recipe probe (gentler lr, shared)	slows, does not cure	shared adapter is structurally interference-bound
yemba per-language probe	2.3 h single-word data vs conversational eval	data-limited; training cannot fix; data-first track
kinyarwanda probe (300 h)	clean U-curve, best 0.375 @ 1800 steps	trainable; scaled decisive run justified
kinyarwanda decisive (300 h, rank 32, 6k steps)	plateau 0.355; FROZEN-POOL 0.372/0.089 vs 0.411/0.087	shipped v1 by owner order at 9.3% rel. gain
kinyarwanda v2 full FT on ~2,000 h	PENDING (data uploading)	ship only if > 9.3% rel. gain

Evidence: platform/evidence/: B5-T6-GATE-2026-001, B5-DIAG-2026-001, B5-SWEEP-2026-001, B5-PROBE-2026-001/-002, B5-KW-DECISIVE-2026-001, B5-CAMPAIGN-R2-TERMINAL-2026-001; SageMaker job medzen-b5-b5-kinyarwanda-full-2026-001 status Completed (listed 2026-08-20).

11. Significant problems, root causes, resolutions

#	Problem	Root cause → resolution
1	Eval attempt 25 failed at container start	asset cards resolve absolute /models paths; pod mounted /input only → /models subPath mount + render-
2	T6 gate box failed 7× (each fail-closed, <\$1 each)	fact-key fetches (no ListBucket) · ECR needs SG self-trust · image ENTRYPOINT → --entrypoint python
3	Adapter merges wrong / rank mismatch	wrap_lora targeted llama_decoder by default → CTC_SCOPE_PREFIX='encoder.'; hardcoded rank 16 vs
4	English ingest box died repeatedly	IAM propagation (wait + verify creds INSIDE box), missing KMS decrypt on repo tarball, ProfileNotFound (
5	Full CV kinyarwanda pull refused at scale	identical audio bytes with different transcripts → explicit opt-in MEDZEN_BYTE_CONFLICT_POLICY=drop
6	Bedrock ValidationException on model id	bare Haiku id needs provisioned throughput → eu.anthropic.... inference profile (config only)
7	Gateway image builds failed	service-dir build context vs repo-root COPYs → repo-root context; Dockerfiles enforce SOURCE_COMMIT
8	Fish secret name wrong in code+runbook+image	medzen/fish-api-key does not exist → default now medzen
9	NV watcher died at its own 10 h ceiling	ingest itself healthy; re-armed with 30 h horizon — lesson: watcher timeouts must exceed the watched pro

12. Architecture / implementation decisions Claude changed, and why

- **Shared multilingual adapter → per-language adapters** (B5-TRAINING-STRATEGY-2026-001): T6 + forensics showed cross-language interference is structural, not a tuning artifact. French/swahili/lingala already meet bars untrained (base model serves them); yemba/ewe/pulaar are data-limited → data-first.
- **LoRA-only → LoRA + full fine-tune mode**: owner chose option B (full FT on all ~2,000 h kinyarwanda) after LoRA plateaued at 0.355 dev; v1 shipped now, v2 must earn its promotion (>9.3%).
- **kinyarwanda bar 0.30 → 0.38 abs + 0.09 rel**: owner-revised on decisive-run evidence; verbatim quotes preserved in the gate file.
- **Mock-only gateways → real providers behind env gates** (owner order): fail-closed defaults kept; fake modes remain the default everywhere.
- **Eval-only Tier B data → training-unlocked** on the owner's private signed agreements, recorded as attestation-only decision records (LIC-2026-003/-004) — the yemba_egra precedent.
- **Byte-dup hard refusal → explicit opt-in drop_both** at scale (never train on contested labels; counts logged for review).

13. AWS resource inventory (all eu-central-1, account 558069890522; verified 2026-08-20)

S3 — single data bucket **medzen-speech** (SSE-KMS key 9c336116-c648-4548-95c6-1b926478ae57; any new role needs kms:Decrypt + GenerateDataKey):

Prefix	Contents
approved/kinyarwanda/asr/v1/model.pt	the shipped v1 model (1.95 GB)
curated/<lang>/asr/<config>/v1/	24 language prefixes; per-corpus audio + A3 manifests (e.g. nv_hausa, cv17_rw, av_pcm pending)
eval/<language>/	FROZEN evaluation pools (40+ languages) — never write here; training data must never byte-match the
raw/_incoming/africanvoices/	owner-uploaded Pidgin archives (7/42 landed at report time; box consumes + deletes)
research/b5-training/	job workspaces, eval bundles, box logs (rw-ingest/, av-ingest/, t6-eval-r2/)
curated/_versions/	dataset generation (gbN) version bindings

ECR images (speech platform; digest prefixes):

Repository:tag	Digest	Role
medzen-orchestrator:v1-2026-08-20a	1d970e6f...	B6 orchestrator
medzen-rag-index:v1-2026-08-20a	bc882db3...	RAG index service
medzen-llm-gateway:v1-bedrock-2026-08-20a	cfc677dd...	LLM gateway (Bedrock)
medzen-speech-tts-gateway:v1-fish-2026-08-20d	46cd5abd...	TTS gateway — SUPERSEDES -20c (bad secret default)
medzen-asr-runtime:b6a-003c-89f94d3	47d86776...	ASR runtime
medzen-model-loader:b6a-003b-69bdce1	ac6f694b...	weights initContainer
medzen-asr-eval-runtime:pilot-8f63996	7f9938e0...	eval harness (gate runs pin sha256:dafc4b14...)
medzen-trainer-omniasr:ft-2026-08-19a	fb15770e...	trainer (LoRA + full-FT modes)

Compute / other:

Resource	Id / name	State & purpose
EC2 (running)	i-04ad7c0f8cb06e9db c6i.4xlarge	rw 2,000 h ingest; self-terminates on completion
EC2 (running)	i-03cd4aa5fd4ff1519 c6i.4xlarge + 1 TB gp3	Pidgin staged ingest; self-terminates; 40 h staging deadline
EKS	cluster medzen-speech; ASGs eks-cpu 0/0/4, eks-gpu 0/0/1	control plane running (≈\$74/mo), zero nodes — serving is dark
SageMaker	quota L-56AE9D73 = 1 (increase to 2 PENDING, case c060a648a19e01astgczeneadmeb615-kinyarwanda-full-2026-001	Completed
ECS (hands off)	cluster medzen-tts-dev, service medzen-tts-gateway	OWNER'S LIVE reference TTS — review only, never modify
IAM (ephemeral)	medzen-av-ingest-ephemeral (role + instance profile)	tear down after Pidgin ingest completes
SSM	/medzen/registry/tts/voices; /medzen/registry/test/b6/<sha>/*	TTS voice registry (kinyarwanda approved, pidgin eval-only); B6 test registry
Secrets	medzen/client-api-keys (HAS VALUE); medzen/speech/fish-api-key (ESWABY auth, TTS first load); medzen/tts/dev/fish-api-key (live dev's own	
Bedrock	eu.anthropic.claude-haiku-4-5-20251001-v1:0	LLM model (eu inference profile — bare ids are refused)
Networking	subnet-01fb2fc3f56bce55e; SGs sg-0fee72d218ac002a7 + sg-03a10ed450a5e20f; MDSP2 private	standard 50a5e20f MDSP2 private 2VPC endpoints, SG self-trust for ECR)

14. Important local directories and files

Path	Contents
~/Documents/medzen-platform/	THE repo (HEAD 6d7d68c). pipeline/ (ingest + 14 adapters + omniasr_train.py), services/ (7 services), scrip
platform/runbooks/B6-DEPLOY-RUNBOOK.md	THE deploy procedure + 2026-08-20 addendum: images, env contract, owner MANUAL ACTION for the Fis
registry/gates/*.yaml	per-language promotion bars; kinyarwanda-v1.yaml carries the owner's verbatim overrides
~/Documents/medzen-shared/codex_status.txt	(585 lines) visible engineering logs; append status here; authorization phrases for training jobs live here
~/Documents/medzen-dropbox/africanvoices/	42 Pidgin archives (154 GB) — the Mac-side source being uploaded; keep until S3 copy verified complete
Session scratchpad (/private/tmp/claude-501/...	box scratchpad) scripts, watcher logs (av-upload.log, naijavoices-ingest.log) — session-lived, not durable

15. Cost & running-resource status

Cost Explorer (month-to-date Aug 1–20, lags ~24 h): gross recorded usage ≈ \$50 (ECR \$20, ECS \$11, EC2-other \$8, RDS \$7, S3 \$2.5), fully offset by credits — net ≈ \$0. The two ingest boxes + SageMaker jobs of the last 48 h are not yet fully reflected.

Currently accruing: 2× c6i.4xlarge (≈\$1.36/h combined, on-demand) + 1 TB gp3 (≈\$0.11/h) — both boxes self-terminate; worst case if both run 3 more days ≈ \$110, reported maximum ≤ **\$210** (owner rule: +\$100 on every reported max). EKS control plane ≈ \$74/mo. ECS medzen-tts-dev and an avatar-chatbot GPU ASG (1/1/1) belong to the owner's other workloads — not this project's engineering, not touched.

Upcoming approved-path spend: kinyarwanda v2 full FT ≈ \$70–150 on-demand, reported maximum ≤ **\$250** — launches only behind an owner-authorized packet.

16. Known issues, risks, technical debt, assumptions

- **BLOCKER (owner-only):** medzen/speech/fish-api-key is EMPTY — fish-mode TTS cannot work until the owner loads the key (runbook has the exact command).
- **Owner decision pending:** serving operating shape (always-on / business-hours / on-demand) — blocks task #37.
- akan/serer baselines known-wrong (#35). LLM calibration unrun (#36).
- Orchestrator pins the TEST registry namespace (/medzen/registry/test/b6/) — production namespace is a deliberate, reviewed deploy-day change; do not quietly widen it.
- Quota increase in human review — plan as if 1 slot until it flips.
- Wave-2's owner-approved 13-language list predates the per-language pivot — re-scope before spending.
- pidgin voice in the TTS registry is EVAL-ONLY (approved:false) — the gateway refuses it for production use by design.
- Watchers/loops in the current Claude session die with the session; the boxes and uploads themselves do NOT depend on the session (only the NV Mac-side queue and the AV upload run on the Mac — keep it awake).
- Assumption baked into ingest: usable() bounds 1–30 s drop very long spontaneous Pidgin takes (metadata shows up to 467 s); drop counts are logged per batch — revisit with segmentation if yield is too low.

17. Exact current task and where work stopped

Work stopped mid-arc “**prep NaijaVoices + Pidgin for the next training session**” (owner-approved plan, in execution): the Pidgin adapter is committed and its ingest box is live in its staging loop; NV igbo is uploading from the Mac; the rw 2,000 h re-pull is uploading from its box; the quota request is pending. **Nothing is blocked on engineering** — the next engineering actions all trigger on completions: (a) rw upload completes → curate v2 → gb5 → v2 packet to owner; (b) Pidgin ingest completes → Tier-B review → decision table; (c) quota flips → note the second slot in the packet.

18. How to verify the current system before changing anything

```
cd ~/Documents/medzen-platform && git log --oneline -15 # recent work
.venv/bin/python -m pytest tests/ -q # expect 2202 passed
export AWS_PROFILE=medzen # all commands below
aws s3 ls s3://medzen-speech/approved/kinyarwanda/asr/v1/ # shipped model
aws s3 cp s3://medzen-speech/research/b5-training/rw-ingest/host.log - | tail -20
aws s3 cp s3://medzen-speech/research/b5-training/av-ingest/host.log - | tail -20
aws ec2 describe-instances --filters Name=instance-state-name,Values=running \
--query 'Reservations[].Instances[].InstanceId,InstanceType,Tags'
aws sagemaker list-training-jobs --sort-by CreationTime --sort-order Descending --max-results 5
aws service-quotas list-requested-service-quota-change-history-by-quota \
--service-code sagemaker --quota-code L-56AE9D73 --region eu-central-1
aws ecr describe-images --repository-name medzen-speech-tts-gateway \
--query 'sort_by(imageDetails,&imagePushedAt)[-1].imageTags' # expect -20d
aws secretsmanager describe-secret --secret-id medzen/speech/fish-api-key \
--query 'VersionIdsToStages' # null = still EMPTY (owner must load)
tail -40 ~/Documents/medzen-shared/codex_status.txt # engineering log
```

19. HANDOFF — START HERE

Inspect first (30 minutes): (1) tail of ~/Documents/medzen-shared/codex_status.txt — the log the owner reads; (2) the two box logs (§18 commands) — they decide your first action; (3) registry/gates/kinyarwanda-v1.yaml — the owner's verbatim decisions you must honour; (4) platform/runbooks/B6-DEPLOY-RUNBOOK.md — the deploy contract; (5) run the test suite.

Do NOT redo: the frozen baselines (suite attempts 29–44), the v1 gate, Tier A ingests + review, the five service images (only rebuild on code change, always with SOURCE_COMMIT), the wave-1 forensics — the conclusions are evidence-backed and owner-ratified. Do not touch ECS medzen-tts-dev. Do not commit anything referencing the owner's private signed agreements beyond the existing attestation records.

Highest-priority task: the kinyarwanda v2 chain. When the rw box logs INGEST_EXIT_0: verify the v2 manifest in curated/kinyarwanda/asr/cv17_rw/v2/, curate (drop_both counts, byte-leak strip vs the frozen pool), bind gb5, then draft the v2 full-FT packet and put the numbers to the owner in the shared file. Launch only after the owner's explicit authorization phrase. Ship as v2 only if relative gain > 9.3% — this is a standing owner order.

Second: when the AV box logs INGEST_EXIT_0, run the Tier-B review (NV hausa/igbo/yoruba + Pidgin), then present the next-wave decision table (bars + costs) to the owner. Tear down the medzen-av-ingest-ephemeral role/profile afterwards.

Owner-only items to surface, not perform: load the Fish key into medzen/speech/fish-api-key; decide the serving operating shape; approve training numbers.

Working discipline that kept this project safe: verify from inside the box before trusting it; fail closed on anything ambiguous; one authorization = one numbered attempt; every gate result becomes a committed evidence record; report costs on-demand +\$100 on maxima; and when a claim matters, re-verify it live — that is how §11 item 8 was caught the day before it would have broken the deploy.